

Is Hybrid Reasoning the Next Big Thing in Artificial Intelligence?

By combining the structure of symbolic AI with the flexibility of statistical AI, hybrid reasoning can reduce hallucinations and help AI systems give more accurate and trustworthy answers.



detectors blocked emails from certain domains. These were entirely symbolic, logic-driven systems.

The machine learning era followed, moving beyond fixed rules. Instead of predetermined logic, systems learned from data, inferring patterns and relationships through statistics, probability, and regression models. This enabled working with unstructured data and discovering patterns rules couldn't capture.

The last fifteen years brought in the deep learning era. Scientists asked: "How can machines think more like humans?" They created artificial neural networks inspired by brain architecture, featuring multiple connection layers—hence 'deep' learning, referring to layered processing.

This evolution has led to today's large language models and the emergence of hybrid reasoning.

Demystifying large language models

Large language models (LLMs) became household terms through platforms like ChatGPT and Claude. But what exactly is an LLM?

I think of an LLM as a massive statistical engine trained on vast internet text—from GitHub to Wikipedia. It learns patterns in word and phrase relationships. In the LLM world, these building blocks are tokens—the smallest text units, like words or word parts.

For example, if I ask an AI system to fill in this blank, "Bengaluru is a __," it examines all contexts where 'Bengaluru' appears in training data. It determines

We are surrounded by AI today. New terms constantly emerge, many fading as quickly as they appear. However, occasionally, a concept emerges that deserves our attention. In my opinion, hybrid reasoning is one such notion.

When I first encountered this term, I questioned its true meaning. Was it merely another piece of jargon? The deeper I explored, the more I realised it connects directly to how humans think and make decisions daily.

Understanding human reasoning through examples

Let me illustrate this with a scenario. You're at work performing routine tasks when your manager approaches: "There's an urgent issue. Can you fix it?" Without hesitation, you agree and resolve it. This quick and almost automatic response represents one type of reasoning.

Now consider another scenario: It's late Friday evening, and your boss presents a new task—a request from the CEO. You pause, weigh options, and carefully choose your response. This slower, deliberate thought process represents another reasoning type.

Hybrid reasoning combines these approaches: fast, rule-based responses with slower, statistical, probability-driven reasoning. When applied to artificial intelligence, something powerful emerges.

The evolution of artificial intelligence

Before exploring hybrid reasoning further, I believe understanding AI's evolution provides crucial context.

Artificial intelligence has existed since the 1950s. Early systems were rule-based, operating on fixed logic. Credit card fraud detection flagged transactions not meeting specific conditions. Spam